🏥 HIMS Backend — Complete REST API Catalog

Comprehensive Endpoint Directory & Schema Reference (45 Total Endpoints)

Production / Testing Host
`https://3.111.31.60.nip.io`

Live Server Base URLs:
- **Secure HTTPS (Recommended for Frontend):** `https://3.111.31.60.nip.io` (SSL via Caddy + nip.io)
- **Direct HTTP (Port 3000 for Postman / Curl):** `http://3.111.31.60:3000`
- **Payloads:** `Content-Type: application/json` • All dates follow `YYYY-MM-DD` • Timestamps follow ISO 8601.

0. SYSTEM & HEALTH PROBES (2 APIS)

METHOD	ENDPOINT	PURPOSE	EXPECTED OUTPUT
GET	`/health`	Service liveness and healthcheck probe	`{"status":"ok","timestamp":"...","service":"HIMS Backend API"}`
GET	`/api`	API Directory & active module catalog	JSON listing all active route namespaces and version info

1. PATIENT MANAGEMENT SERVICE (6 APIS) — `/API/PATIENTS`

Handles patient demographic records, search indexing, and deterministic UHID generation.

POST `/api/patients` AUTO UHID GENERATION

Register new patient. The PostgreSQL trigger automatically assigns next sequential UHID (e.g. `HIMS-2026-00001`).

```
POST /api/patients
Content-Type: application/json

{
  "first_name": "Rohan",
  "last_name": "Sharma",
  "date_of_birth": "1994-08-12",
  "gender": "male",                // 'male' | 'female' | 'other'
  "blood_group": "O+",             // 'A+' | 'A-' | 'B+' | 'B-' | 'AB+' | 'AB-' | 'O+' | 'O-'
  "phone": "+919876500001",
  "email": "rohan.sharma@example.com",
  "address": "Flat 302, Green Glen Layout, Bangalore",
  "emergency_contact_name": "Sunita Sharma",
  "emergency_contact_phone": "+919876500002"
}

// Response (201 Created):
{
  "id": "c0000000-0000-0000-0000-000000000001",
  "uhid": "HIMS-2026-00001",
  "first_name": "Rohan",
  "last_name": "Sharma",
  "phone": "+919876500001",
  "is_active": true,
  "created_at": "2026-10-27T10:00:00.000Z"
}
```

GET `/api/patients` SEARCH & PAGINATION

Query: `?search=<name_or_phone>&page=1&limit=10`. Searches both first_name and last_name case-insensitively.

```
// Response (200 OK):
{
  "patients": [ { "id": "...", "uhid": "HIMS-2026-00001", ... } ],
  "pagination": { "page": 1, "limit": 10, "total": 1, "total_pages": 1 }
}
```

GET `/api/patients/:id`

Get patient by internal UUID (e.g. `/api/patients/c0000000-0000-0000-0000-000000000001`).

GET `/api/patients/uhid/:uhid`

Lookup patient by hospital UHID (e.g. `/api/patients/uhid/HIMS-2026-00001`).

PATCH **/api/patients/:id**

Update patient address, phone, or emergency contacts. UHID is immutable.

DELETE **/api/patients/:id**

Soft-delete patient record (marks `is_active = false`).

2. OPD & APPOINTMENTS QUEUE SERVICE (11 APIS) — `/API/OPD`

Orchestrates departments, doctors, scheduling, status lifecycles, and live token queue display.

GET **`/api/opd/departments`**
List all active hospital clinical departments.

POST **`/api/opd/departments`**
Create department: `{"name": "Cardiology", "code": "CARD"}`. Code must be unique.

GET **`/api/opd/doctors`** `FILTER BY DEPT`
Query: `?departmentId=<UUID>&available=true`. Returns active doctors and their specializations.

POST **`/api/opd/doctors`**
Register doctor: `{"user_id": "<UUID>", "full_name": "Dr. Arvind", "specialization": "Cardiologist", "department_id": "<UUID>"}`.

GET **`/api/opd/doctors/:id`** | **`PATCH /api/opd/doctors/:id`**
Get doctor details or update availability status (`{"is_available": false}`).

POST **`/api/opd/appointments`** `AUTO DAILY TOKEN`
Book appointment. The database auto-assigns next sequential token number for that department & date.

```
POST /api/opd/appointments
Content-Type: application/json

{
  "patient_id": "c0000000-0000-0000-0000-000000000001",
  "doctor_id": "a0000000-0000-0000-0000-000000000001",
  "department_id": "d0000000-0000-0000-0000-000000000001",
  "scheduled_at": "2026-10-27T10:30:00.000Z",
  "notes": "Follow-up for chest discomfort"
}

// Response (201 Created):
{
  "id": "e0000000-0000-0000-0000-000000000001",
  "token_number": 1,
  "status": "scheduled",
  "scheduled_at": "2026-10-27T10:30:00.000Z"
}
```

GET **`/api/opd/appointments`**
Query: `?patientId=&doctorId=&deptId=&date=YYYY-MM-DD&status=&page=1&limit=20`.

GET **`/api/opd/appointments/:id`**
Get complete appointment details including patient and doctor info.

PATCH **`/api/opd/appointments/:id/status`**
Status transitions: `scheduled` → `checked_in` → `in_progress` → `completed` (or `cancelled` / `no_show`).

GET **`/api/opd/appointments/queue/:departmentId/:date`** `WAITING ROOM DISPLAY`
Returns ordered patient queue for a department on a date (`YYYY-MM-DD`). Perfect for waiting room TVs.

3. CLINICAL & EMR SERVICE (8 APIS) — `/API/CLINICAL`

Provides physician consultation logging, ICD-10 diagnostic coding, digital prescriptions, and referrals.

POST **`/api/clinical`** `FULL CONSULTATION`
Create consultation with embedded prescription items and department referrals in one call.

```
POST /api/clinical
Content-Type: application/json

{
  "patient_id": "c0000000-0000-0000-0000-000000000001",
  "appointment_id": "e0000000-0000-0000-0000-000000000001",
  "doctor_id": "a0000000-0000-0000-0000-000000000001",
  "chief_complaint": "Chest pain and breathlessness",
  "history_of_illness": "Started 3 days ago.",
  "examination_notes": "BP: 130/85, Pulse: 78 bpm.",
  "diagnosis": "Non-cardiac chest pain, rule out angina",
  "icd10_codes": ["R07.9", "R06.0"],
  "follow_up_date": "2026-11-05",
  "status": "completed",
  "prescriptions": [
    { "drug_name": "Aspirin 75mg", "dosage": "75mg", "frequency": "Once daily after dinner", "duration_days": 14, "instructions":
"Take after meals" }
  ],
  "referrals": [
    { "referred_to_dept": "Radiology", "reason": "Chest X-Ray and ECG", "priority": "routine" }
  ]
}
```

GET **`/api/clinical | /api/clinical/:id`**
List consultations (filter by `?patientId=&doctorId=&status=`) or fetch single consultation with all sub-items.

GET **`/api/clinical/appointment/:appointmentId`**
Fetch 1:1 consultation tied to an appointment.

PATCH **`/api/clinical/:id`**
Update diagnosis, notes, follow-up date, or status (`open` → `completed` → `referred`).

POST **`/api/clinical/:id/prescriptions`**
Add additional prescription items: `{"items": [{drug_name, dosage, frequency, duration_days, instructions}]}`.

PATCH **`/api/clinical/prescriptions/:itemId/dispensed`**
Mark prescription item as dispensed (`{"dispensed": true}`).

POST **`/api/clinical/:id/referrals`**
Add cross-department referral: `{"referred_to_dept": "Neurology", "reason": "Headache", "priority": "urgent"}`.

4. LABORATORY INFORMATION SYSTEM (LIMS) (9 APIS) — `/API/LIMS`

Manages diagnostic test catalogs, specimen orders, numerical result entry, and secure PDF cloud storage.

GET **`/api/lims/tests`** | **`POST /api/lims/tests`**
List lab test catalog or create test: `{"test_name": "CBC", "test_code": "CBC", "category": "Haematology", "price": 350}`.

POST **`/api/lims/orders`**
Create lab order with items: `{"patient_id": "...", "ordered_by": "...", "priority": "routine", "items": [{"test_id": "..."}]}`.

GET **`/api/lims/orders`** | **`/api/lims/orders/:id`**
List orders (filter by `?patientId=&status=&priority=`) or fetch order details with test results.

PATCH **`/api/lims/orders/:id/status`**
Update workflow stage: `pending` → `sample_collected` → `processing` → `resulted` → `delivered`.

PATCH **`/api/lims/orders/items/:itemId/result`**
Enter findings: `{"result_value": "14.2", "result_unit": "g/dL", "reference_range": "13.5-17.5", "is_abnormal": false}`.

POST **`/api/lims/orders/:id/report`**
Attach uploaded report path: `{"report_path": "lab-reports/2026/10/order-1.pdf", "report_file_name": "CBC.pdf", "report_uploaded_by": "<UUID>"}`.

GET **`/api/lims/orders/:id/report/download`** `SIGNED URL`
Generates a secure 5-minute pre-signed download URL from Supabase Storage bucket `lab-reports`.

5. PHARMACY & INVENTORY MODULE (9 APIS) — `/API/PHARMACY`

Formulary catalog, stock batches with expiry dates, low-stock alerts, and atomic FIFO dispensing.

GET **`/api/pharmacy/drugs`** | **`/api/pharmacy/drugs/alerts/low-stock`**
List medications with live stock count or get drugs currently below reorder threshold.

GET **`/api/pharmacy/drugs/:id`**
Get drug details along with all active stock batches and expiry dates.

POST **`/api/pharmacy/drugs`** | **`PATCH /api/pharmacy/drugs/:id`**
Add drug (`{name, unit, price_per_unit, reorder_level}`) or update price/reorder level.

POST **`/api/pharmacy/stock`**
Receive new inventory batch: `{"drug_id": "...", "batch_number": "BAT-2026-999", "expiry_date": "2027-12-31", "quantity": 500}`.

POST **`/api/pharmacy/dispense`** `ATOMIC FIFO DEDUCT`
ACID Dispense. Calls PL/pgSQL `dispense_drugs` function. Deducts from batches FIFO by earliest expiry using row locks.

```
POST /api/pharmacy/dispense
Content-Type: application/json

{
  "patient_id": "c0000000-0000-0000-0000-000000000001",
  "dispensed_by": "b0000000-0000-0000-0000-000000000001",
  "items": [
    { "drug_id": "dd000000-0000-0000-0000-000000000001", "drug_name": "Paracetamol 500mg", "quantity": 10 }
  ]
}
```

GET **`/api/pharmacy/dispense`** | **`/api/pharmacy/dispense/:id`**
List dispense orders history or view detailed batch breakdown of a specific dispense transaction.

6. BILLING & INVOICING SERVICE (8 APIS) — `/API/FINANCE`

Point-of-Sale billing, multi-department line item charges, multi-tender payment recording, and balance calculation.

POST **`/api/finance`** `CREATE INVOICE`
Create draft invoice for patient visit with optional initial line items.

```
POST /api/finance
Content-Type: application/json

{
  "patient_id": "c0000000-0000-0000-0000-000000000001",
  "visit_date": "2026-10-27",
  "discount_amount": 50.00,
  "tax_amount": 25.00,
  "line_items": [
    { "charge_type": "consultation", "description": "Cardiology Consultation", "quantity": 1, "unit_price": 500.00 }
  ]
}
```

POST **/api/finance/:id/items** ADD CHARGES

Attach items: `{"items": [{charge_type: "lab"|"pharmacy"|"consultation"|"procedure", description, quantity, unit_price}]}`.

POST **/api/finance/:id/payments** RECORD PAYMENT

Record payment (`cash`, `card`, `upi`, `insurance`, `waiver`). Auto-updates invoice status to `partially_paid` or `paid`.

```
POST /api/finance/ff000000-0000-0000-0000-000000000001/payments
Content-Type: application/json

{
  "amount": 475.00,
  "method": "upi",
  "reference_no": "UPI-TXN-2026102712345",
  "collected_by": "b0000000-0000-0000-0000-000000000001"
}
```

GET **/api/finance/:id**

Full invoice record with patient info, line items, payments, and real-time SQL balance summary.

GET **/api/finance/summary/patient/:patientId** PATIENT FINANCIAL SUMMARY

Queries PostgreSQL `invoice_summary` view. Returns subtotal, discount, tax, total, paid, and balance due for all visits.

PATCH **/api/finance/:id/issue | /api/finance/:id/cancel**

Issue invoice (locks final total) or cancel invoice.

7. APPENDIX: TEST UUIDS FOR INSTANT POSTMAN / FRONTEND TESTING

DOMAIN	ENTITY	UUID / CODE	DETAILS
Patient	Rohan Sharma	`c0000000-0000-0000-0000-000000000001`	UHID: `HIMS-2026-00001` • Phone: `+919876500001`
Patient	Priya Patel	`c0000000-0000-0000-0000-000000000002`	UHID: `HIMS-2026-00002` • Phone: `+919876500002`
OPD Dept	Cardiology	`d0000000-0000-0000-0000-000000000001`	Code: `CARD`
OPD Dept	Pediatrics	`d0000000-0000-0000-0000-000000000002`	Code: `PED`
Doctor	Dr. Arvind Swaminathan	`a0000000-0000-0000-0000-000000000001`	Cardiology • MD
Doctor	Dr. Sunita Rao	`a0000000-0000-0000-0000-000000000002`	Pediatrics • MBBS, DCH
LIMS Test	Complete Blood Count	`aa000000-0000-0000-0000-000000000001`	Code: `CBC` • Price: ₹350.00
LIMS Test	Lipid Profile	`aa000000-0000-0000-0000-000000000003`	Code: `LIPID` • Price: ₹750.00
Pharmacy	Paracetamol 500mg	`dd000000-0000-0000-0000-000000000001`	Unit: tablet • Price: ₹4.00
Pharmacy	Amoxicillin 500mg	`dd000000-0000-0000-0000-000000000002`	Unit: capsule • Price: ₹12.50
Staff User	Admin / Cashier	`b0000000-0000-0000-0000-000000000001`	Collector ID for payments & dispensing